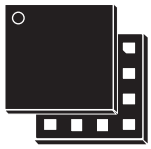
LIS3MDL

Digital output magnetic sensor: ultra-low-power, high-performance 3-axis magnetometer

Datasheet - production data



LGA-12 (2.0x2.0x1.0 mm)

Features

- Wide supply voltage, 1.9 V to 3.6 V
- Independent IO supply (1.8 V)
- ±4/ ±8/ ±12/ ±16 gauss selectable magnetic full scale
- Continuous and single-conversion modes
- 16-bit data output
- Interrupt generator
- Self-test
- I^2C/SPI digital output interface
- Power-down mode/ low-power mode
- ECOPACK®, RoHS and “Green” compliant

Applications

- Magnetometers
- Compasses

Description

The LIS3MDL is an ultra-low-power high-performance three-axis magnetic sensor.

The LIS3MDL has user-selectable full scales of ±4/ ±8/ ±12/ ±16 gauss.

The self-test capability allows the user to check the functioning of the sensor in the final application.

The device may be configured to generate interrupt signals for magnetic field detection.

The LIS3MDL includes an I^2C serial bus interface that supports standard and fast mode (100 kHz and 400 kHz) and SPI serial standard interface.

The LIS3MDL is available in a small thin plastic land grid array package (LGA) and is guaranteed to operate over an extended temperature range of -40 °C to +85 °C.

Table 1. Device summary

Order codes	Temperature range [°C]	Package	Packaging
LIS3MDL	-40 to +85	LGA-12	Tray
LIS3MDLTR	-40 to +85	LGA-12	Tape and reel

This is information on a product in full production. www.st.com

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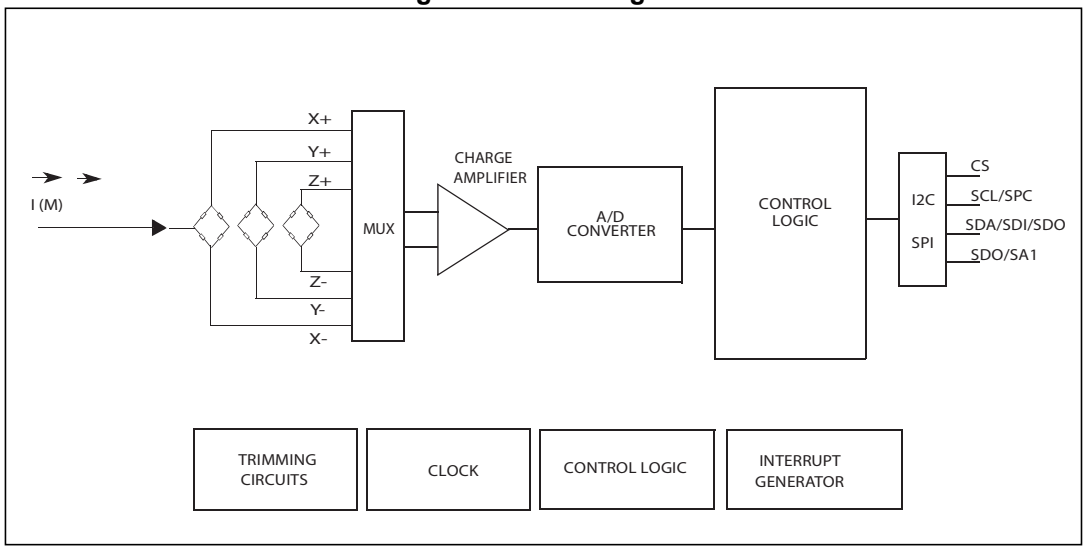
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1 Block diagram and pin description

1.1 Block diagram

Figure 1. Block diagram



1.2 Pin description

Figure 2. Pin connections

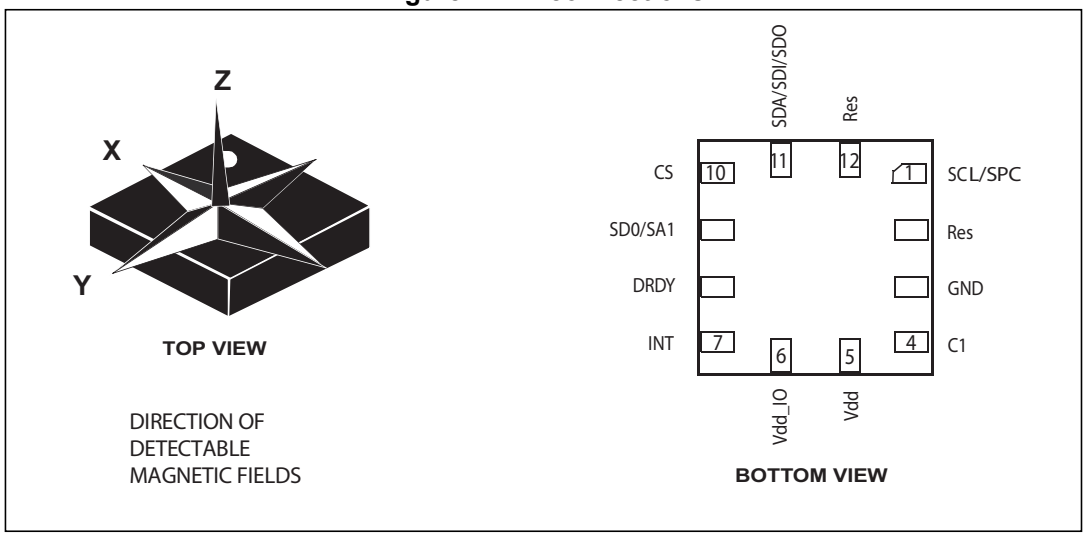


Table 2. Pin description

Pin#	Name	Function
1	SCL SPC	I^2C serial clock (SCL) SPI serial port clock (SPC)
2	Reserved	Connect to GND
3	GND	Connect to GND
4	C1	Capacitor connection (C1=100 nF)
5	Vdd	Power supply
6	Vdd_IO	Power supply for I/O pins
7	INT	Interrupt
8	DRDY	Data Ready
9	SDO SA1	SPI serial data output (SDO) I^2C less significant bit of the device address (SA1)
10	CS	SPI enable I^2C/SPI mode selection (1: SPI idle mode / I^2C communication enabled; 0: SPI communication mode / I^2C disabled)
11	SDA SDI SDO	I^2C serial data (SDA) SPI serial data input (SDI) 3-wire interface serial data output (SDO)
12	Reserved	Connect to GND

2 Magnetic and electrical specifications

2.1 Magnetic characteristics

@ Vdd = 2.5 V, T = 25 °C unless otherwise noted[a]

Table 3. Mechanical characteristics

Symbol	Parameter	Test conditions	Min.	Typ.[1]	Max.	Unit
FS	Measurement range			±4		gauss
				±8		
				±12		
				±16		
GN	Sensitivity	FS=±4 gauss		6842		LSB/ gauss
		FS=±8 gauss		3421		
		FS=±12 gauss		2281		
		FS=±16 gauss		1711		
Zgauss	Zero-gauss level	FS=±4 gauss		±1		gauss
RMS	RMS noise	X-axis; FS=±12 gauss; Ultra-high-performance mode		3.2		mgauss
		Y-axis; FS=±12 gauss Ultra-high-performance mode		3.2		mgauss
		Z-axis; FS=±12 gauss Ultra-high-performance mode		4.1		mgauss
NL	Non-linearity	Best-fit straight line FS = ±12 gauss Happlied = ±6 gauss		±0.12		%FS
ST	Self test[2]	X-axis FS = ±12 gauss	1		3	gauss
		Y-axis FS = ±12 gauss	1		3	
		Z-axis FS = ±12 gauss	0.1		1	
DF	Magnetic disturbance field	Zero-gauss offset starts to degrade			50	gauss
Top	Operating temperature range		-40		+85	°C

1. Typical specifications are not guaranteed.
2. Absolute value.

a. The product is factory calibrated at 2.5 V. The operational power supply range is from 1.9 V to 3.6 V.

2.2 Temperature sensor characteristics

@ Vdd = 2.5 V, T = 25 °C unless otherwise noted[b].

Table 4. Temperature sensor characteristics

Symbol	Parameter	Test conditions	Min.	Typ.[1]	Max.	Unit
TSDr	Temperature sensor output change vs. temperature	-		8		LSB/°C
TODR	Temperature refresh rate[2]			ODR		Hz
Top	Operating temperature range		-40		+85	°C

1. Typical specifications are not guaranteed.
2. If TEMP_EN bit in *CTRL_REG1 (20h)* is set to'1', a temperature data is acquired at each conversion cycle. Refer to *Table 22*.

2.3 Electrical characteristics

@ Vdd = 2.5 V, T = 25 °C unless otherwise noted[c]

Table 5. Electrical characteristics

Symbol	Parameter	Test conditions	Min.	Typ.[1]	Max.	Unit
Vdd	Supply voltage		1.9		3.6	V
Vdd_IO	Power supply for I/O		1.71	1.8	Vdd+0.1	
Idd_HR	Current consumption in ultra-high-resolution mode	ODR = 20 Hz		270		µA
Idd_LP	Current consumption in low-power mode	ODR = 20 Hz		40		µA
Idd_PD	Current consumption in power down			1		µA
Top	Operating temperature range		-40		+85	°C

1. Typical specification are not guaranteed.

b. The product is factory calibrated at 2.5 V.

c. The product is factory calibrated at 2.5 V. The operational power supply range is from 1.9 V to 3.6 V.

2.4 Communication interface characteristics

2.4.1 SPI - serial peripheral interface

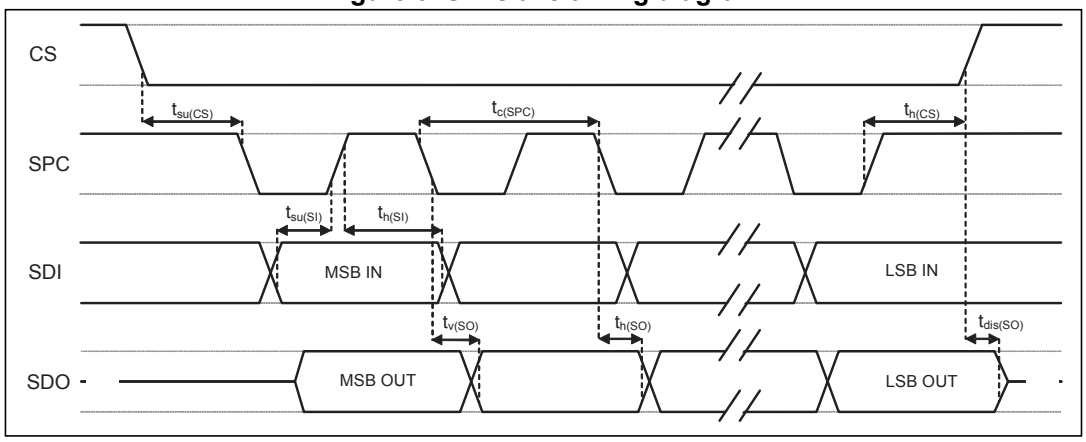
Subject to general operating conditions for Vdd and Top.

Table 6. SPI slave timing values

Symbol	Parameter	Value (1)		Unit
		Min.	Max.	
$t_{c(SPC)}$	SPI clock cycle	100		ns
$f_{c(SPC)}$	SPI clock frequency		10	MHz
$t_{su(CS)}$	CS setup time	5		ns
$t_{h(CS)}$	CS hold time	20		
$t_{su(SI)}$	SDI input setup time	5		
$t_{h(SI)}$	SDI input hold time	15		
$t_{v(SO)}$	SDO valid output time		50	
$t_{h(SO)}$	SDO output hold time	5		
$t_{dis(SO)}$	SDO output disable time		50	

1. Values are guaranteed at 10 MHz clock frequency for SPI with both 4 and 3 wires, based on characterization results, not tested in production.

Figure 3. SPI slave timing diagram



Note: Measurement points are done at 0.2• Vdd_IO and 0.8• Vdd_IO, for both input and output ports.

2.4.2 Sensor I²C - inter IC control interface

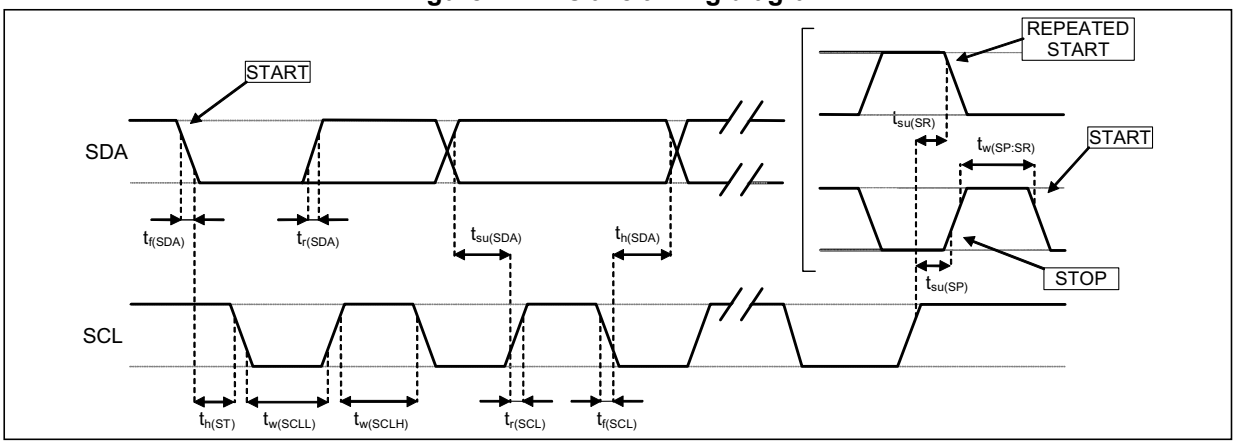
Subject to general operating conditions for Vdd and Top.

Table 7. I²C slave timing values

Symbol	Parameter	I²C standard mode (1)		I²C fast mode (1)		Unit
		Min.	Max.	Min.	Max.	
$f_{(SCL)}$	SCL clock frequency	0	100	0	400	kHz
$t_{w(SCLL)}$	SCL clock low time	4.7		1.3		µs
$t_{w(SCLH)}$	SCL clock high time	4.0		0.6		
$t_{su(SDA)}$	SDA setup time	250		100		ns
$t_{h(SDA)}$	SDA data hold time	0	3.45	0	0.9	µs
$t_{r(SDA)}$ $t_{r(SCL)}$	SDA and SCL rise time		1000	$20 + 0.1C_b$ (2)	300	ns
$t_{f(SDA)}$ $t_{f(SCL)}$	SDA and SCL fall time		300	$20 + 0.1C_b$ (2)	300	
$t_{h(ST)}$	START condition hold time	4		0.6		µs
$t_{su(SR)}$	Repeated START condition setup time	4.7		0.6		
$t_{su(SP)}$	STOP condition setup time	4		0.6		
$t_{w(SP:SR)}$	Bus free time between STOP and START condition	4.7		1.3		

1. Data based on standard I²C protocol requirement, not tested in production.
2. C_b = total capacitance of one bus line, in pF.

Figure 4. I²C slave timing diagram



Note: Measurement points are done at 0.2• Vdd_IO and 0.8• Vdd_IO, for both ports.

2.5 Absolute maximum ratings

Stresses above those listed as “absolute maximum ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device under these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Table 8. Absolute maximum ratings

Symbol	Ratings	Maximum value	Unit
Vdd	Supply voltage	-0.3 to 4.8	V
Vdd_IO	I/O pins supply voltage	-0.3 to 4.8	V
Vin	Input voltage on any control pin (SCL/SPC, SDA/SDI/SDO, SDO/SA1, CS)	-0.3 to Vdd_IO +0.3	V
A_{UNP}	Acceleration (any axis)	3,000 for 0.5 ms	g
		10,000 for 0.1 ms	g
M_{EF}	Maximum exposed field	1000	gauss
T_{OP}	Operating temperature range	-40 to +85	°C
T_{STG}	Storage temperature range	-40 to +125	°C

Note: Supply voltage on any pin should never exceed 4.8 V.



This device is sensitive to magnetic fields, improper handling can cause permanent damage to the part.



This device is sensitive to electrostatic discharge (ESD), improper handling can cause permanent damage to the part.

3 Terminology and functionality

3.1 Sensitivity

Sensitivity describes the gain of the sensor and can be determined, for example, by applying a magnetic field of 1 *gauss* to it.

3.2 Zero-*gauss* level

Zero-*g*auss level offset describes the deviation of an actual output signal from the ideal output if no magnetic field is present.

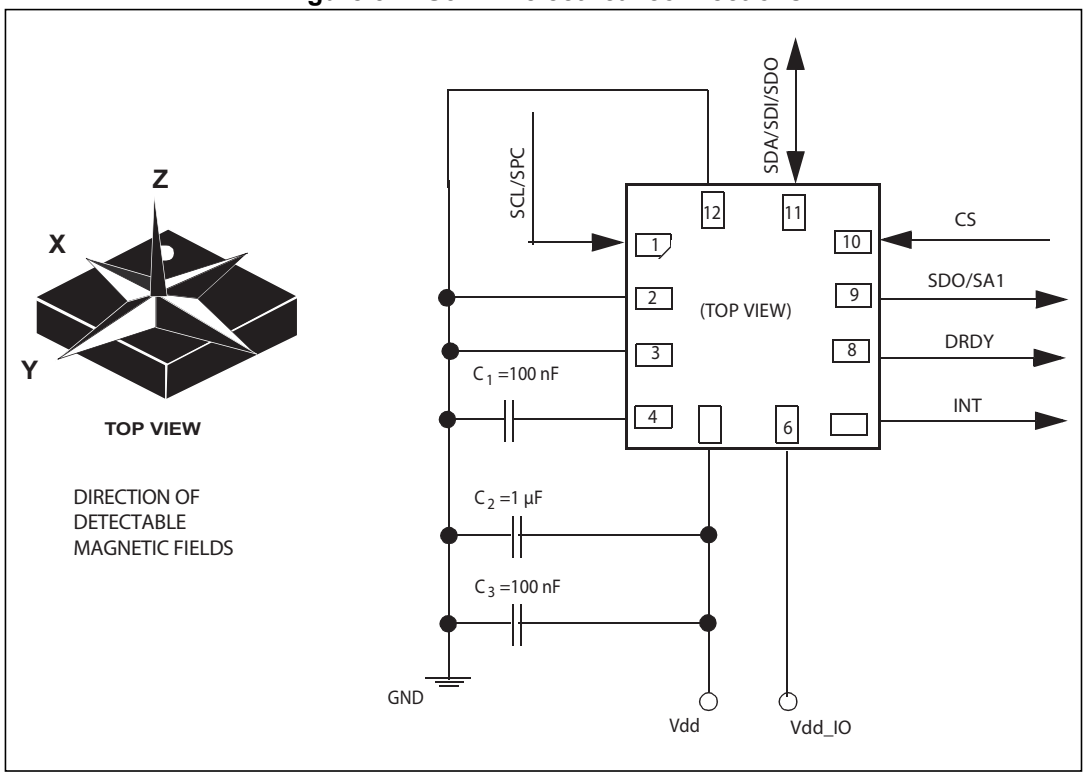
3.3 Factory calibration

The IC interface is factory calibrated for sensitivity (So) and Zero-*gauss* level (TyOff).

The trimming values are stored in the device in non-volatile memory. Each time the device is turned on, the trimming parameters are downloaded to the registers to be employed during active operation which allows using the device without further calibration.

4 Application hints

Figure 5. LIS3MDL electrical connections



4.1 External capacitors

The LIS3MDL requires one external capacitor (C_1 = 100 *nF)* connected between pin 4 and GND.

The device core power supply line (Vdd) needs one high-frequency decoupling capacitor (C_3 = 100 *nF,* ceramic) as near as possible to the supply pin, and a low-frequency electrolytic capacitor (C_2 = 1 *µF)*. All the voltage and ground supplies must be present at the same time to have proper behavior of the IC (refer to *Figure 5*).

The functionality of the device and the measured magnetic field data is selectable and accessible through the I^2C / SPI interfaces.

The functions, the threshold and the timing of the interrupt pin (INT) can be completely programmed by the user through the I^2C / SPI interfaces.

When I^2C or 3-wire SPI is used, the SDO/SA1 pin must be connected to Vdd_IO or GND.

4.2 Soldering information

The LGA package is compliant with the ECOPACK®, RoHS and “Green” standard.
It is qualified for soldering heat resistance according to JEDEC J-STD-020.

Land pattern and soldering recommendations are available at www.st.com.

4.3 High-current wiring effects

High current in wiring and printed circuit traces can cause errors in magnetic field measurements for compassing.

Conductor-generated magnetic fields will add to the Earth's magnetic field, causing errors in compass heading computation.

Keep currents higher than 10 mA a few millimeters away from the sensor IC.

5 Digital interfaces

The registers embedded in the LIS3MDL may be accessed through both the I^2C and SPI serial interfaces. The latter may be SW configured to operate either in 3-wire or 4-wire interface mode.

The serial interfaces are mapped to the same pads. To select/exploit the I^2C interface, the CS line must be tied high (i.e. connected to Vdd_IO).

Table 9. Serial interface pin description

Pin name	Pin description
CS	SPI enable I^2C/SPI mode selection (1: SPI idle mode / I^2C communication enabled; 0: SPI communication mode / I^2C disabled)
SCL SPC	I^2C serial clock (SCL) SPI serial port clock (SPC)
SDA SDI SDO	I^2C serial data (SDA) SPI serial data input (SDI) 3-wire interface serial data output (SDO)
SA1 SDO	I^2C less significant bit of the device address (SA1) SPI serial data output (SDO)

5.1 I^2C serial interface

The LIS3MDL I^2C is a bus slave. The I^2C is employed to write data to registers whose content can also be read back.

The relevant I^2C terminology is given in the table below.

Table 10. I^2C terminology

Term	Description
Transmitter	The device which sends data to the bus
Receiver	The device which receives data from the bus
Master	The device which initiates a transfer, generates clock signals and terminates a transfer
Slave	The device addressed by the master

There are two signals associated with the I^2C bus: the serial clock line (SCL) and the serial data line (SDA). The latter is a bidirectional line used for sending and receiving the data to/from the interface. Both lines must be connected to Vdd_IO through an external pull-up resistor. When the bus is free, both the lines are high.

The I^2C interface is compliant with fast mode (400 kHz) I^2C standards, as well as with normal mode.

When the I^2C interface is used, the SDO/SA1 pin has to be connected to Vdd_IO or GND.

5.1.1 I²C operation

The transaction on the bus is started through a START (ST) signal. A START condition is defined as a HIGH-to-LOW transition on the data line while the SCL line is held HIGH. After this has been transmitted by the master, the bus is considered busy. The next byte of data transmitted after the start condition contains the address of the slave in the first seven bits and the eighth bit tells whether the master is receiving data from the slave or transmitting data to the slave. When an address is sent, each device in the system compares the first seven bits after a start condition with its address. If they match, the device considers itself addressed by the master.

The Slave ADdress (SAD) associated to the LIS3MDL is 00111x0b, whereas the x bit is modified by the SDO/SA1 pin in order to modify the device address. If the SDO/SA1 pin is connected to the voltage supply, the address is 0011110b, otherwise, if the SDO/SA1 pin is connected to ground, the address is 0011100b.

Data transfer with acknowledge is mandatory. The transmitter must release the SDA line during the acknowledge pulse. The receiver must then pull the data line LOW so that it remains stable low during the HIGH period of the acknowledge clock pulse. A receiver which has been addressed is obliged to generate an acknowledge after each byte of data received.

The I²C embedded inside the LIS3MDL behaves like a slave device and the following protocol must be adhered to. After the START condition (ST) a slave address is sent, once a slave acknowledge (SAK) has been returned, an 8-bit sub-address (SUB) is transmitted: the 7 LSb represent the actual register address while the MSb enables address auto-increment. If the MSb of the SUB field is ‘1’, the SUB (register address) is automatically increased to allow multiple data read/write.

The slave address is completed with a Read/Write bit. If the bit is ‘1’ (Read), a repeated START (SR) condition must be issued after the two sub-address bytes; if the bit is ‘0’ (Write) the master will transmit to the slave with direction unchanged. *Table 11* explains how the SAD+read/write bit pattern is composed, listing all the possible configurations.

Table 11. SAD+read/write patterns

Command	SAD[6:2]	SAD[1] = SDO/SA1	SAD[0]	R/W	SAD+R/W
Read	00111	0	0	1	00111001 (39h)
Write	00111	0	0	0	00111000 (38h)
Read	00111	1	0	1	00111101 (3Dh)
Write	00111	1	0	0	00111100 (3Ch)

Table 12. Transfer when master is writing one byte to slave

Master	ST	SAD + W		SUB		DATA		SP
Slave			SAK		SAK		SAK	

Table 13. Transfer when master is writing multiple bytes to slave

Master	ST	SAD + W		SUB		DATA		DATA		SP
Slave			SAK		SAK		SAK		SAK	

Table 14. Transfer when master is receiving (reading) one byte of data from slave

Master	ST	SAD + W		SUB		SR	SAD + R			NMAK	SP
Slave			SAK		SAK			SAK	DATA		

Table 15. Transfer when master is receiving (reading) multiple bytes of data from slave

Master	ST	SAD+W		SUB		SR	SAD+R			MAK		MAK		NMAK	SP
Slave			SAK		SAK			SAK	DATA		DAT A		DATA		

Data are transmitted in byte format (DATA). Each data transfer contains 8 bits. The number of bytes transferred per transfer is unlimited. Data is transferred with the most significant bit (MSb) first. If a receiver cannot receive another complete byte of data until it has performed some other function, it can hold the clock line SCL LOW to force the transmitter into a wait state. Data transfer only continues when the receiver is ready for another byte and releases the data line. If a slave receiver does not acknowledge the slave address (i.e. it is not able to receive because it is performing some real-time function) the data line must be left HIGH by the slave. The Master can then abort the transfer. A LOW to HIGH transition on the SDA line while the SCL line is HIGH is defined as a STOP condition. Each data transfer must be terminated by the generation of a STOP (SP) condition.

In order to read multiple bytes, it is necessary to assert the most significant bit of the sub-address field. In other words, SUB(7) must be equal to 1, while SUB(6-0) represents the address of first register to be read.

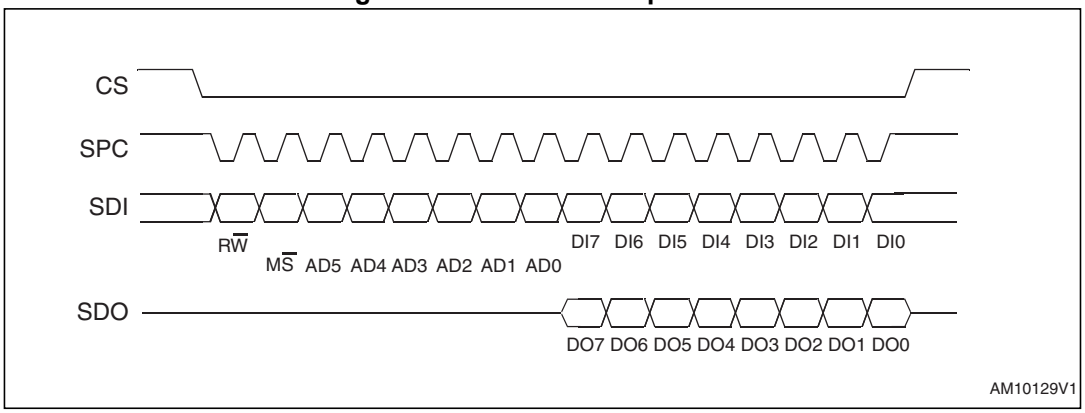
In the presented communication format, MAK is Master acknowledge and NMAK is No Master Acknowledge.

5.2 SPI bus interface

The LIS3MDL SPI is a bus slave. The SPI allows writing to and reading from the registers of the device.

The serial interface interacts with the application through 4 wires: **CS**, **SPC**, **SDI** and **SDO**.

Figure 6. Read and write protocol



CS is the serial port enable and it is controlled by the SPI master. It goes low at the start of the transmission and goes back high at the end. **SPC** is the serial port clock and it is controlled by the SPI master. It is stopped high when **CS** is high (no transmission). **SDI** and **SDO** are respectively the serial port data input and output. Those lines are driven at the falling edge of **SPC** and should be captured at the rising edge of **SPC**.

Both the read register and write register commands are completed in 16 clock pulses or in multiples of 8 in case of multiple byte read/write. Bit duration is the time between two falling edges of **SPC**. The first bit (bit 0) starts at the first falling edge of **SPC** after the falling edge of **CS** while the last bit (bit 15, bit 23, ...) starts at the last falling edge of SPC just before the rising edge of **CS**.

bit 0: R$\overline{W}$ bit. When 0, the data DI(7:0) is written into the device. When 1, the data DO(7:0) from the device is read. In latter case, the chip will drive **SDO** at the start of bit 8.

bit 1: M$\overline{S}$ bit. When 0, the address will remain unchanged in multiple read/write commands. When 1, the address is auto-incremented in multiple read/write commands.

bit 2-7: address AD(5:0). This is the address field of the indexed register.

bit 8-15: data DI(7:0) (write mode). This is the data that is written into the device (MSb first).

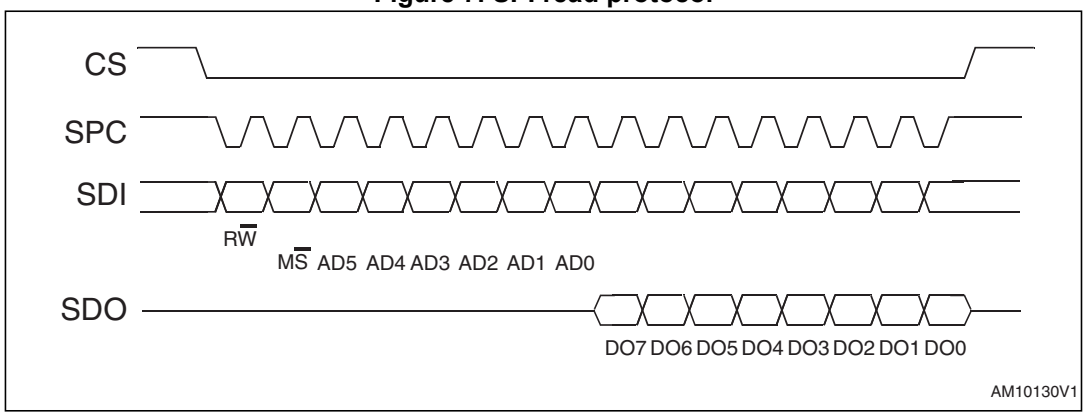
bit 8-15: data DO(7:0) (read mode). This is the data that is read from the device (MSb first).

In multiple read/write commands further blocks of 8 clock periods will be added. When the M$\overline{S}$ bit is ‘0’, the address used to read/write data remains the same for every block. When the M$\overline{S}$ bit is ‘1’, the address used to read/write data is increased at every block.

The function and the behavior of **SDI** and **SDO** remain unchanged.

5.2.1 SPI read

Figure 7. SPI read protocol



The SPI read command is performed with 16 clock pulses. A multiple byte read command is performed by adding blocks of 8 clock pulses to the previous one.

bit 0: READ bit. The value is 1.

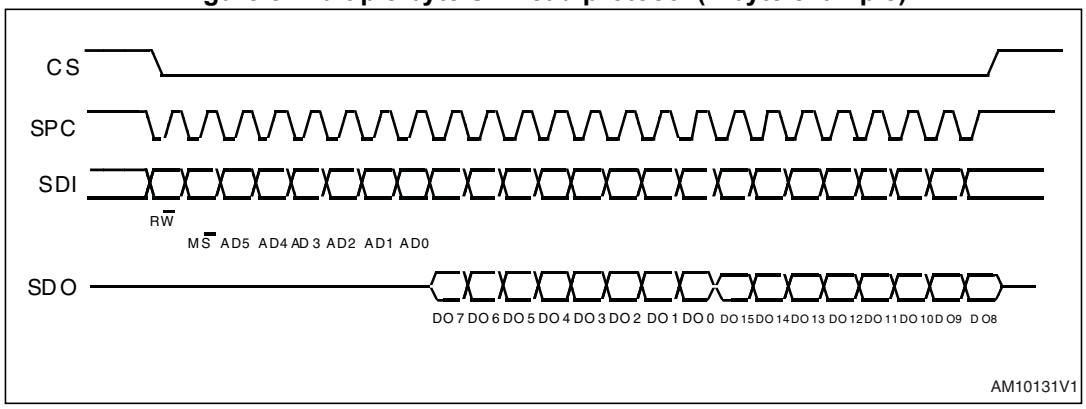
bit 1: MS bit. When 0, does not increment address; when 1, increments the address in multiple reads.

bit 2-7: address AD(5:0). This is the address field of the indexed register.

bit 8-15: data DO(7:0) (read mode). This is the data that will be read from the device (MSb first).

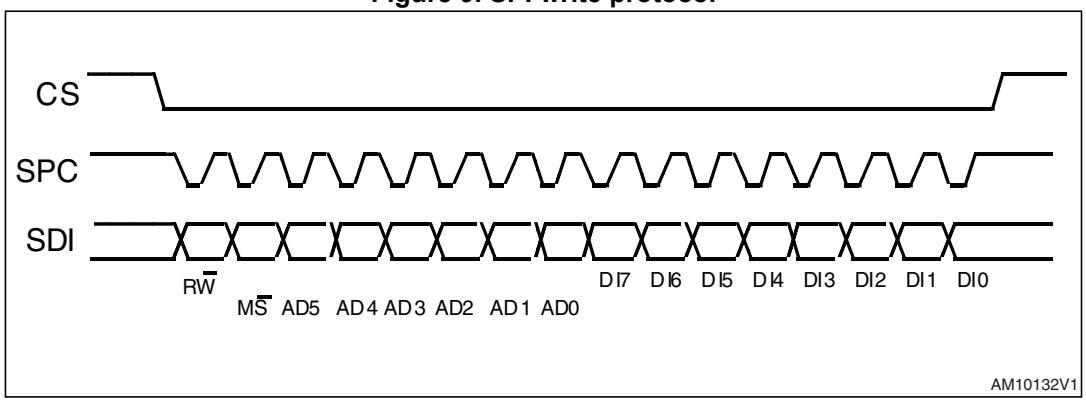
bit 16-... : data DO(...-8). Further data in multiple byte reads.

Figure 8. Multiple byte SPI read protocol (2-byte example)



5.2.2 SPI write

Figure 9. SPI write protocol



The SPI Write command is performed with 16 clock pulses. A multiple byte write command is performed by adding blocks of 8 clock pulses to the previous one.

bit 0: WRITE bit. The value is 0.

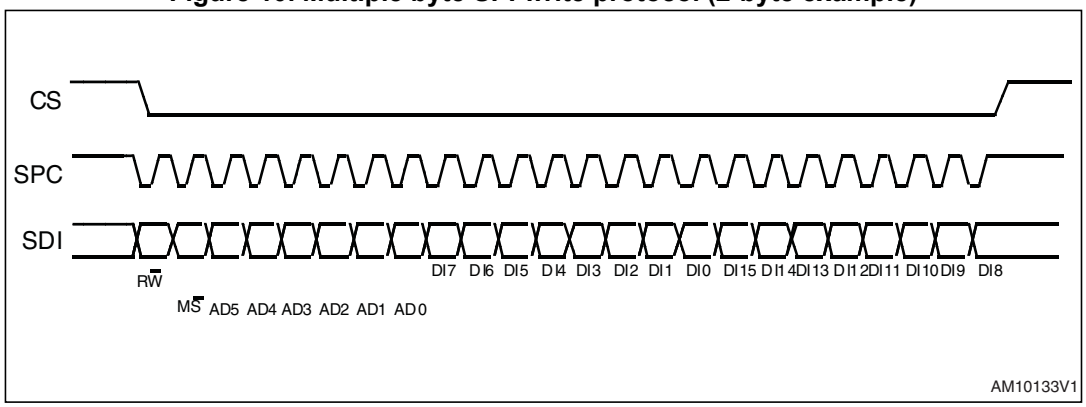
bit 1: MS bit. When 0, does not increment the address; when 1, increments the address in multiple writes.

bit 2 -7: address AD(5:0). This is the address field of the indexed register.

bit 8-15: data DI(7:0) (write mode). This is the data that is written inside the device (MSb first).

bit 16-... : data DI(...-8). Further data in multiple byte writes.

Figure 10. Multiple byte SPI write protocol (2-byte example)

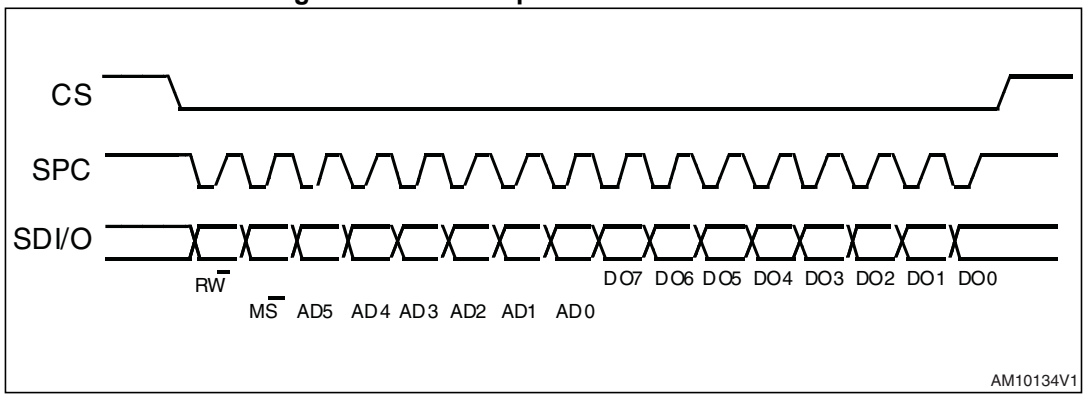


5.2.3 SPI read in 3-wire mode

3-wire mode is entered by setting bit SIM to '1' (SPI serial interface mode selection) in *CTRL_REG3 (22h)*.

When 3-wire mode is used, the SDO/SA1 pin has to be connected to GND or Vdd_IO.

Figure 11. SPI read protocol in 3-wire mode



The SPI read command is performed with 16 clock pulses:

bit 0: READ bit. The value is 1.

bit 1: $M\overline{S}$ bit. When 0, does not increment the address; when 1, increments the address in multiple reads.

bit 2-7: address AD(5:0). This is the address field of the indexed register.

bit 8-15: data DO(7:0) (read mode). This is the data that is read from the device (MSb first).

A multiple read command is also available in 3-wire mode.

6 Register mapping

The table below provides a list of the 8-bit registers embedded in the device and their respective addresses.

Table 16. Register address map

Name	Type	Register address		Default	Comment
		Hex	Binary		
Reserved		00 - 0E	--	--	Reserved
WHO_AM_I	r	0F	0000 1111	00111101	Dummy register
Reserved		10 - 1F	--	--	Reserved
CTRL_REG1	r/w	20	0010 0000	00010000	
CTRL_REG2	r/w	21	0010 0001	00000000	
CTRL_REG3	r/w	22	0010 0010	00000011	
CTRL_REG4	r/w	23	0010 0011	00000000	
CTRL_REG5	r/w	24	0010 0100	00000000	
Reserved		25 - 26	--	--	Reserved
STATUS_REG	r	27	0010 0111	Output	
OUT_X_L	r	28	0010 1000	Output	
OUT_X_H	r	29	0010 1001	Output	
OUT_Y_L	r	2A	0010 1010	Output	
OUT_Y_H	r	2B	0010 1011	Output	
OUT_Z_L	r	2C	0010 1100	Output	
OUT_Z_H	r	2D	0010 1101	Output	
TEMP_OUT_L	r	2E	0010 1110	Output	
TEMP_OUT_H	r	2F	0010 1111	Output	
INT_CFG	rw	30	00110000	00000000	
INT_SRC	r	31	00110001	00000000	
INT_THS_L	r	32	00110010	00000000	
INT_THS_H	r	33	00110011	00000000	

Registers marked *Reserved* or not listed in the table above must not be changed. Writing to those registers may cause permanent damage to the device.

The content of the registers that are loaded at boot should not be changed. They contain the factory calibration values. Their content is automatically restored when the device is powered up.

7 Registers description

7.1 WHO_AM_I (0Fh)

Table 17. WHO_AM_I register

0	0	1	1	1	1	0	1

Device identification register.

7.2 CTRL_REG1 (20h)

Table 18. CTRL_REG1 register

TEMP_EN	OM1	OM0	DO2	DO1	DO0	FAST_ODR	ST

Table 19. CTRL_REG1 description

TEMP_EN	Temperature sensor enable. Default value: 0 (0: temperature sensor disabled; 1: temperature sensor enabled)
OM[1:0]	X and Y axes operative mode selection. Default value: 00 (Refer to *Table 21*)
DO[2:0]	Output data rate selection. Default value: 100 (Refer to *Table 22*)
FAST_ODR	FAST_ODR enables data rates higher than 80 Hz (refer to *Table 20*). Default value: 0 (0: Fast_ODR disabled; 1: FAST_ODR enabled)
ST	Self-test enable. Default value: 0 (0: self-test disabled; 1: self-test enabled)

Table 20. Data rate configuration

DO2	DO1	DO0	FAST_ODR	ODR [Hz]	OM
X	X	X	1	1000	LP
X	X	X	1	560	MP
X	X	X	1	300	HP
X	X	X	1	155	UHP

Table 21. X and Y axes operating mode selection

OM1	OM0	Operating mode for X and Y axes
0	0	Low-power mode
0	1	Medium-performance mode
1	0	High-performance mode
1	1	Ultra-high-performance mode

Table 22. Output data rate configuration

DO2	DO1	DO0	ODR [Hz]
0	0	0	0.625
0	0	1	1.25
0	1	0	2.5
0	1	1	5
1	0	0	10
1	0	1	20
1	1	0	40
1	1	1	80

7.3 CTRL_REG2 (21h)

Table 23. CTRL_REG2 register

0(1)	FS1	FS0	0(1)	REBOOT	SOFT_RST	0(1)	0(1)

1. These bits must be set to ‘0’ for correct functioning of the device

Table 24. CTRL_REG2 description

FS[1:0]	Full-scale configuration. Default value: 00 Refer to *Table 25*
REBOOT	Reboot memory content. Default value: 0 (0: normal mode; 1: reboot memory content)
SOFT_RST	Configuration registers and user register reset function. (0: Default value; 1: Reset operation)

Table 25. Full-scale selection

FS1	FS0	Full-scale
0	0	±4 gauss
0	1	±8 gauss
1	0	±12 gauss
1	1	±16 gauss

7.4 CTRL_REG3 (22h)

Table 26. CTRL_REG3 register

0(1)	0(1)	LP	0(1)	0(1)	SIM	MD1	MD0

1. These bits must be set to ‘0’ for correct functioning of the device

Table 27. CTRL_REG3 description

LP	Low-power mode configuration. Default value: 0 If this bit is ‘1’, DO[2:0] is set to 0.625 Hz and the system performs, for each channel, the minimum number of averages. Once the bit is set to ‘0’, the magnetic data rate is configured by the DO bits in *CTRL_REG1 (20h)* register.
SIM	SPI serial interface mode selection. Default value: 0 (0: 4-wire interface; 1: 3-wire interface).
MD[1:0]	Operating mode selection. Default value: 11 Refer to *Table 28.*

Table 28. System operating mode selection

MD1	MD0	Mode
0	0	Continuous-conversion mode
0	1	Single-conversion mode Single-conversion mode has to be used with sampling frequency from 0.625 Hz to 80Hz.
1	0	Power-down mode
1	1	Power-down mode

7.5 CTRL_REG4 (23h)

Table 29. CTRL_REG4 register

0(1)	0(1)	0(1)	0(1)	OMZ1	OMZ0	BLE	0(1)

1. These bits must be set to ‘0’ for correct functioning of the device

Table 30. CTRL_REG4 description

OMZ[1:0]	Z-axis operative mode selection. Default value: 00. Refer to *Table 31*.
BLE	Big/Little Endian data selection. Default value: 0 (0: data LSb at lower address; 1: data MSb at lower address)

Table 31. Z-axis operating mode selection

OMZ1	OMZ0	Operating mode for Z-axis
0	0	Low-power mode
0	1	Medium-performance mode
1	0	High-performance mode
1	1	Ultra-high-performance mode

7.6 CTRL_REG5 (24h)

Table 32. CTRL_REG5 register

FAST_READ	BDU	0(1)	0(1)	0(1)	0(1)	0(1)	0(1)

1. These bits must be set to '0' for correct functioning of the device

Table 33. CTRL_REG5 description

FAST_READ	FAST READ allows reading the high part of DATA OUT only in order to increase reading efficiency. Default value: 0 (0: FAST_READ disabled; 1: FAST_READ enabled)
BDU	Block data update for magnetic data. Default value: 0 (0: continuous update; 1: output registers not updated until MSb and LSb have been read)

7.7 STATUS_REG (27h)

Table 34. STATUS_REG register

ZYXOR	ZOR	YOR	XOR	ZYXDA	ZDA	YDA	XDA

Table 35. STATUS_REG description

ZYXOR	X-, Y- and Z-axis data overrun. Default value: 0 (0: no overrun has occurred; 1: a new set of data has overwritten the previous set)
ZOR	Z-axis data overrun. Default value: 0 (0: no overrun has occurred; 1: new data for the Z-axis has overwritten the previous data)
YOR	Y-axis data overrun. Default value: 0 (0: no overrun has occurred; 1: new data for the Y-axis has overwritten the previous data)
XOR	X-axis data overrun. Default value: 0 (0: no overrun has occurred; 1: new data for the X-axis has overwritten the previous data)

Table 35. STATUS_REG description (continued)

ZYXDA	X-, Y- and Z-axis new data available. Default value: 0 (0: a new set of data is not yet available; 1: a new set of data is available)
ZDA	Z-axis new data available. Default value: 0 (0: new data for the Z-axis is not yet available; 1: new data for the Z-axis is available)
YDA	Y-axis new data available. Default value: 0 (0: new data for the Y-axis is not yet available; 1: new data for the Y-axis is available)
XDA	X-axis new data available. Default value: 0 (0: new data for the X-axis is not yet available; 1: new data for the X-axis is available)

7.8 OUT_X_L (28h), OUT_X_H(29h)

X-axis data output. The value of magnetic field is expressed as two's complement.

7.9 OUT_Y_L (2Ah), OUT_Y_H (2Bh)

Y-axis data output. The value of magnetic field is expressed as two's complement.

7.10 OUT_Z_L (2Ch), OUT_Z_H (2Dh)

Z-axis data output. The value of magnetic field is expressed as two's complement.

7.11 TEMP_OUT_L (2Eh), TEMP_OUT_H (2Fh)

Temperature sensor data. The value of temperature is expressed as two's complement.

7.12 INT_CFG (30h)

Table 36. INT_CFG register

XIEN	YIEN	ZIEN	0(1)	0(1)	IEA	LIR	IEN

1. These bits must be set to '0' for correct functioning of the device

Table 37. INT_CFG description

XIEN	Enable interrupt generation on X-axis. Default value: 0 0: disable interrupt request; 1: enable interrupt request
YIEN	Enable interrupt generation on Y-axis. Default value: 0 0: disable interrupt request; 1: enable interrupt request
ZIEN	Enable interrupt generation on Z-axis. Default value: 0 0: disable interrupt request; 1: enable interrupt request
IEA	Interrupt active configuration on INT. Default value 0 0: low; 1:high
LIR	Latch interrupt request. Default value: 0 0: interrupt request latched; 1: interrupt request not latched) Once latched, the INT pin remains in the same state until INT_SRC (31h) is read.
IEN	Interrupt enable on INT pin. Default value 0. 0: disabled; 1: enabled

7.13 INT_SRC (31h)

Table 38. INT_SRC register

PTH_X	PTH_Y	PTH_Z	NTH_X	NTH_Y	NTH_Z	MROI	INT

Table 39. INT_SRC description

PTH_X	Value on X-axis exceeds the threshold on the positive side. Default value: 0.
PTH_Y	Value on Y-axis exceeds the threshold on the positive side. Default value: 0.
PTH_Z	Value on Z-axis exceeds the threshold on the positive side. Default value: 0.
NTH_X	Value on X-axis exceeds the threshold on the negative side. Default value: 0.
NTH_Y	Value on Y-axis exceeds the threshold on the negative side. Default value: 0.
NTH_Z	Value on Z-axis exceeds the threshold on the negative side. Default value: 0.
MROI	Internal measurement range overflow on magnetic value. Default value: 0.
INT	This bit signals when an interrupt event occurs.

7.14 INT_THS_L(32h), INT_THS_H(33h)

Interrupt threshold. Default value: 0.

The value is expressed in 16-bit unsigned.

Even if the threshold is expressed in absolute value, the device detects both positive and negative thresholds.

Table 40. INT_THS_L_M

THS7	THS6	THS5	THS4	THS3	THS2	THS1	THS0

Table 41. INT_THS_H_M

0(1)	THS14	THS13	THS12	THS11	THS10	THS9	THS8

1. These bits must be set to ‘0’ for correct functioning of the device

8 Package information

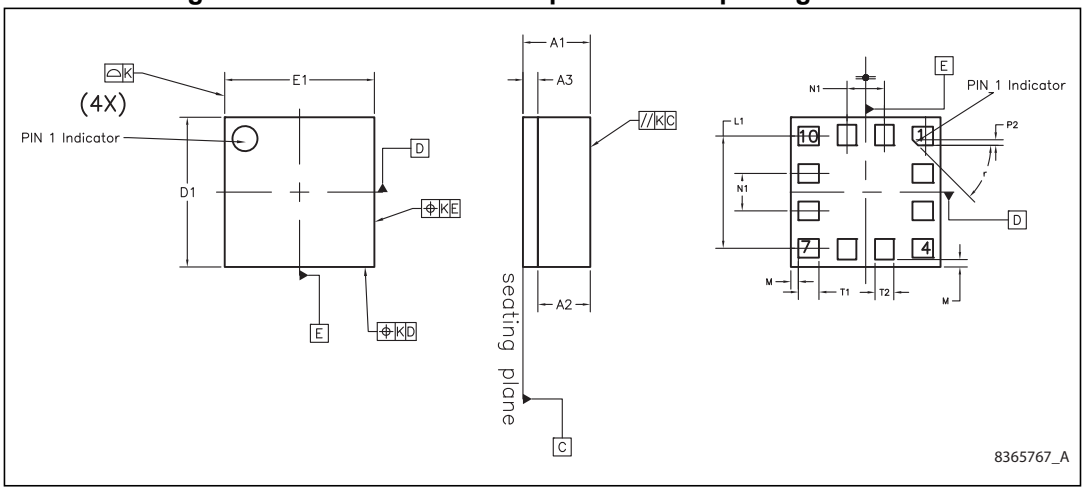
In order to meet environmental requirements, ST offers these devices in different grades of ECOPACK® packages, depending on their level of environmental compliance. ECOPACK® specifications, grade definitions and product status are available at: www.st.com. ECOPACK is an ST trademark.

8.1 VFLGA-12 package information

Table 42. VFLGA 2x2x1 12LD pitch 0.5 mm package mechanical data

Dim.	mm		
	Min.	Typ.	Max.
A1			1
A2		0.785	
A3		0.200	
D1	1.850	2.000	2.150
E1	1.850	2.000	2.150
L1		1.500	
N1		0.500	
T1		0.275	
T2		0.250	
P2		0.075	
r		45°	
M		0.100	
K		0.050	

Figure 12. VFLGA 2x2x1 12LD pitch 0.5 mm package outline



9 Revision history

Table 43. Document revision history

Date	Revision	Changes
01-Feb-2013	1	Initial release
22-Apr-2013	2	Updated note *on page 12* Product status changed from preliminary data to production data
12-Dec-2014	3	Added FAST_ODR bit to *Table 18: CTRL_REG1 register* and *Table 19: CTRL_REG1 description* Added FAST_READ bit to *Table 32: CTRL_REG5 register* and *Table 33: CTRL_REG5 description* Updated *Table 16: Register address map* Minor textual updates throughout document
15-May-2015	4	Added *Table 20: Data rate configuration*

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